

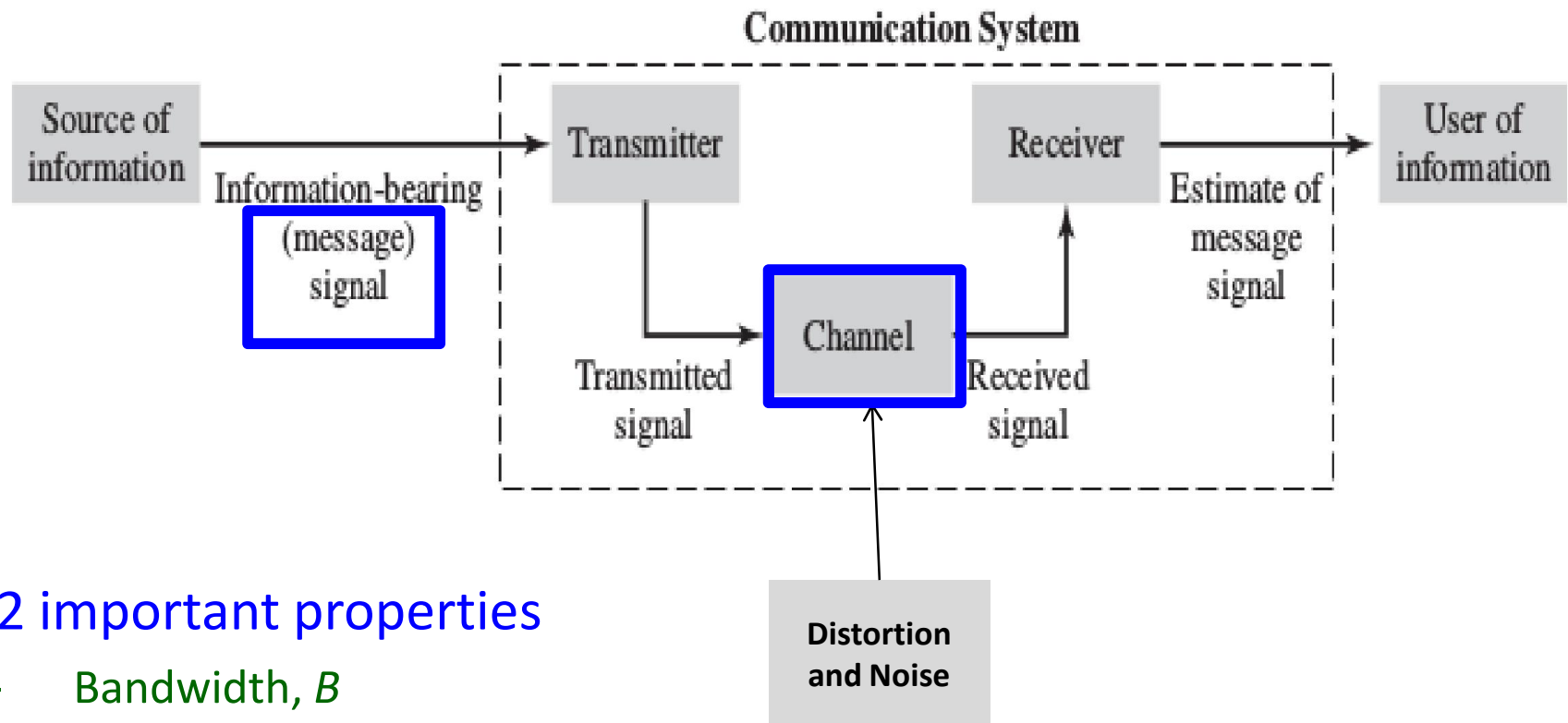
CSE 311:

Data Communication

Instructor:

Dr. Md. Monirul Islam

Channel & signal Characteristics



- 2 important properties
 - Bandwidth, B
 - Signal power, P_s

Channel & signal Characteristics

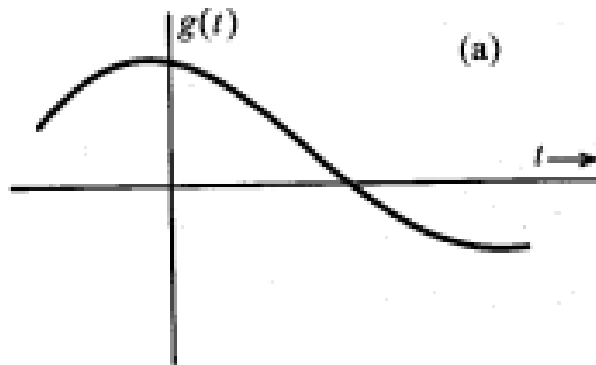
- Bandwidth
 - Channel Bandwidth
 - Signal bandwidth

Channel & signal Characteristics

- Bandwidth
 - Channel Bandwidth
 - Range of frequencies that a channel can transmit
 - Ex: if transmit 0-5 kHz frequencies, Channel bandwidth $B = 5$ kHz
 - Signal bandwidth

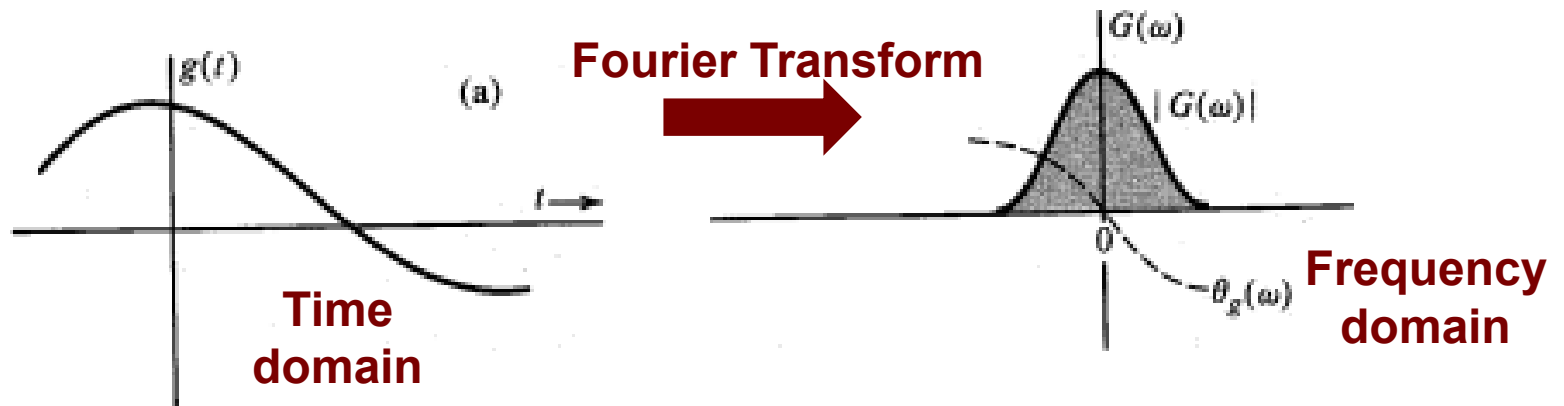
Channel & signal Characteristics

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 - Maximum frequency that is available in a signal



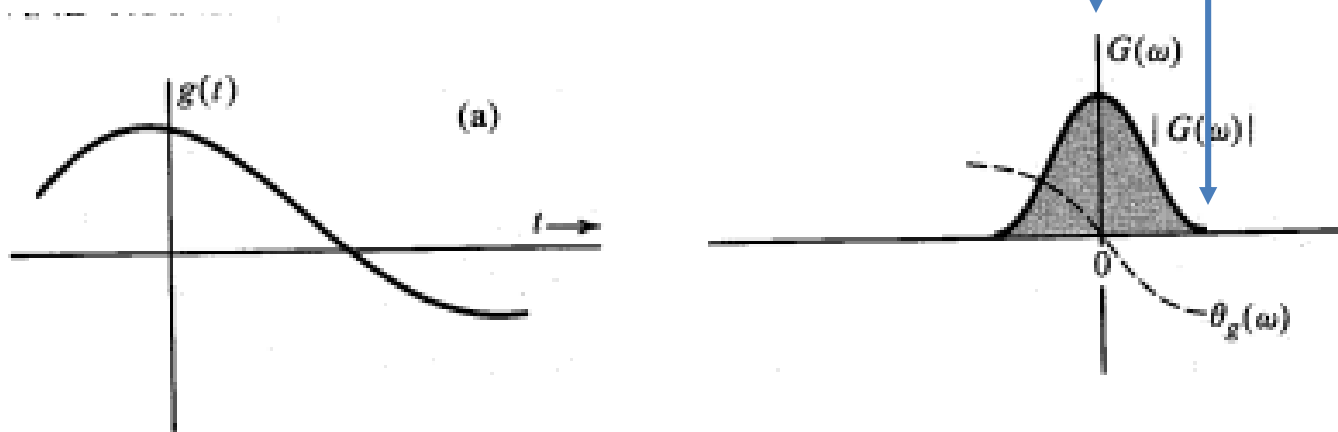
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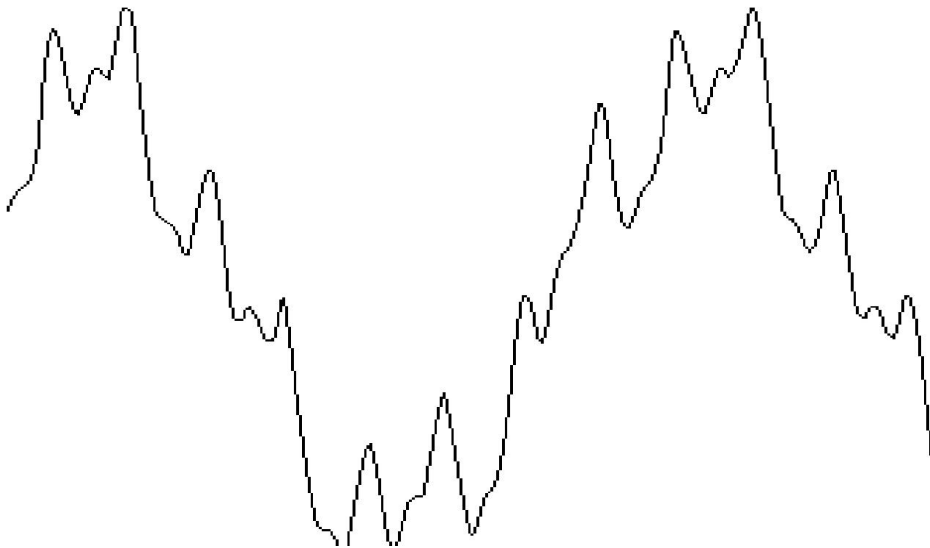


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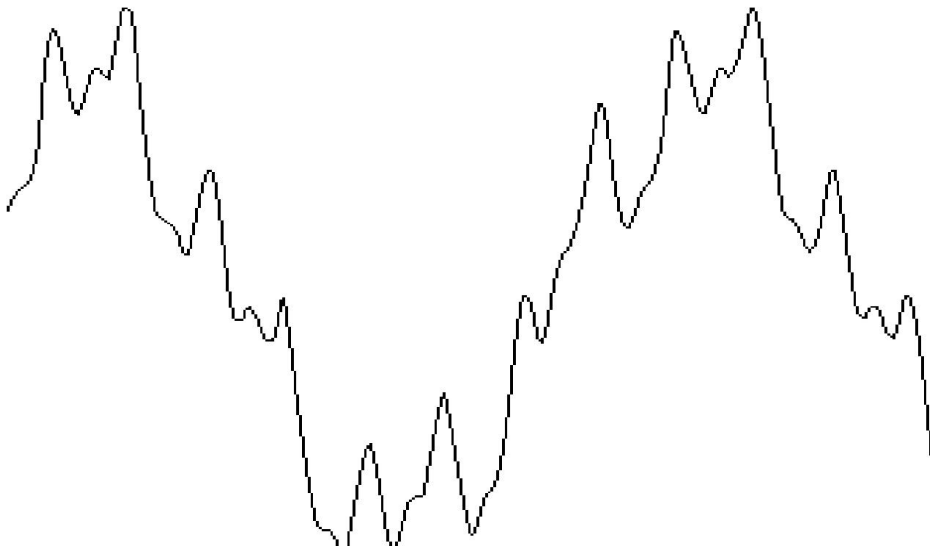


Channel & signal Characteristics

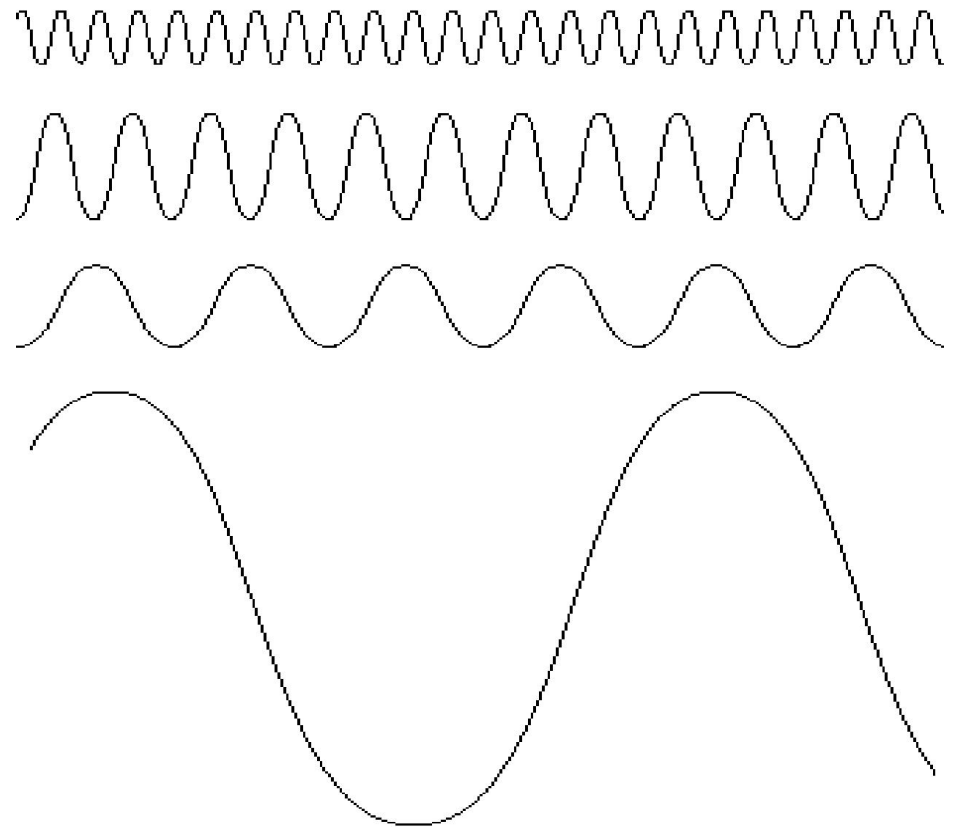


A complex signal with
multiple frequencies

Channel & signal Characteristics



A complex signal with
multiple frequencies



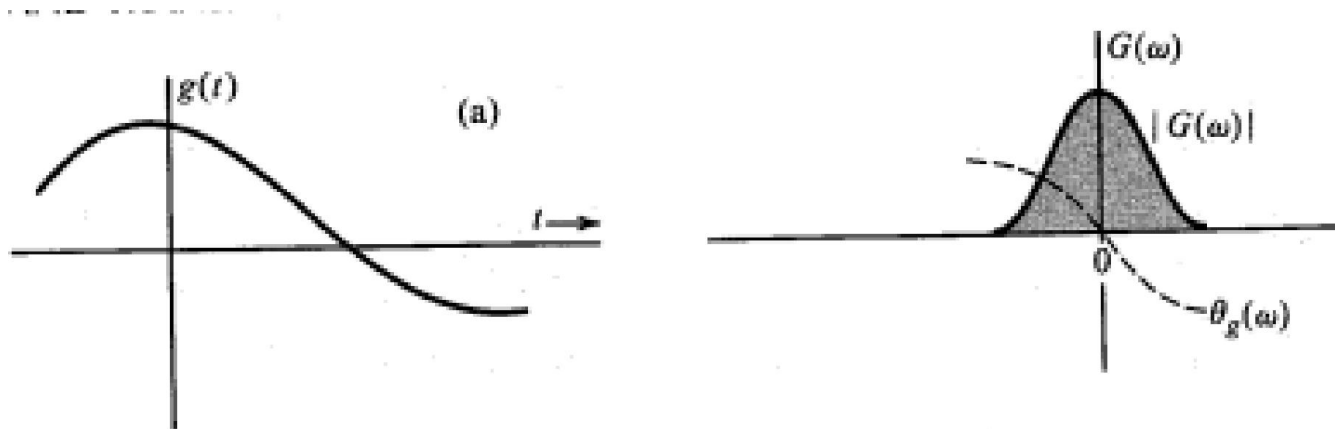
Individual signals each with
a single frequency

Channel & signal Characteristics

- Frequency
 - Change in signal values
 - Faster change in values means higher frequencies
 - High frequency signals
 - Sports/battle scene
 - Low frequency signals
 - News/sleeping animal videos

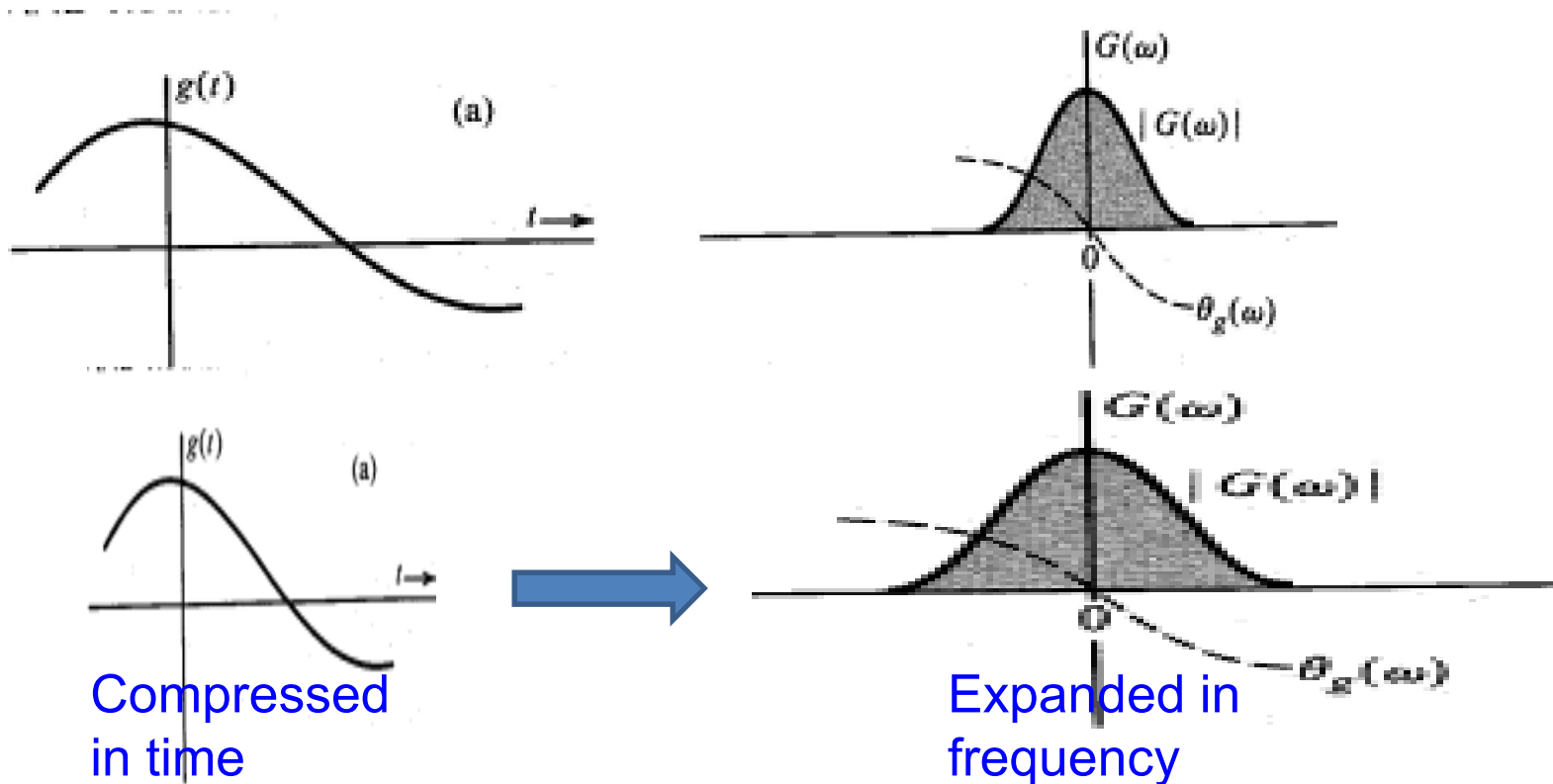
Channel & signal Characteristics

- Frequency
 - Compressing in time, increases frequency, means higher channel Bandwidth



Channel & signal Characteristics

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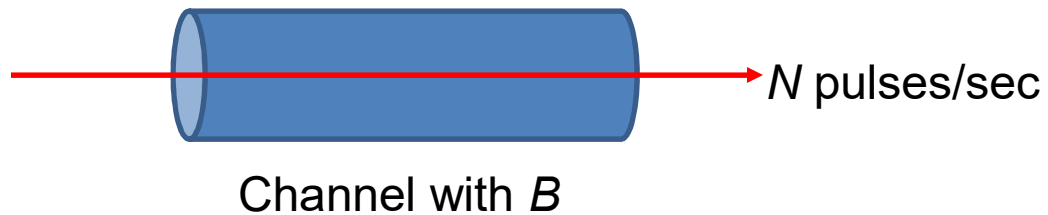


Channel & signal Characteristics

- More transmission speed requires channel with higher bandwidth

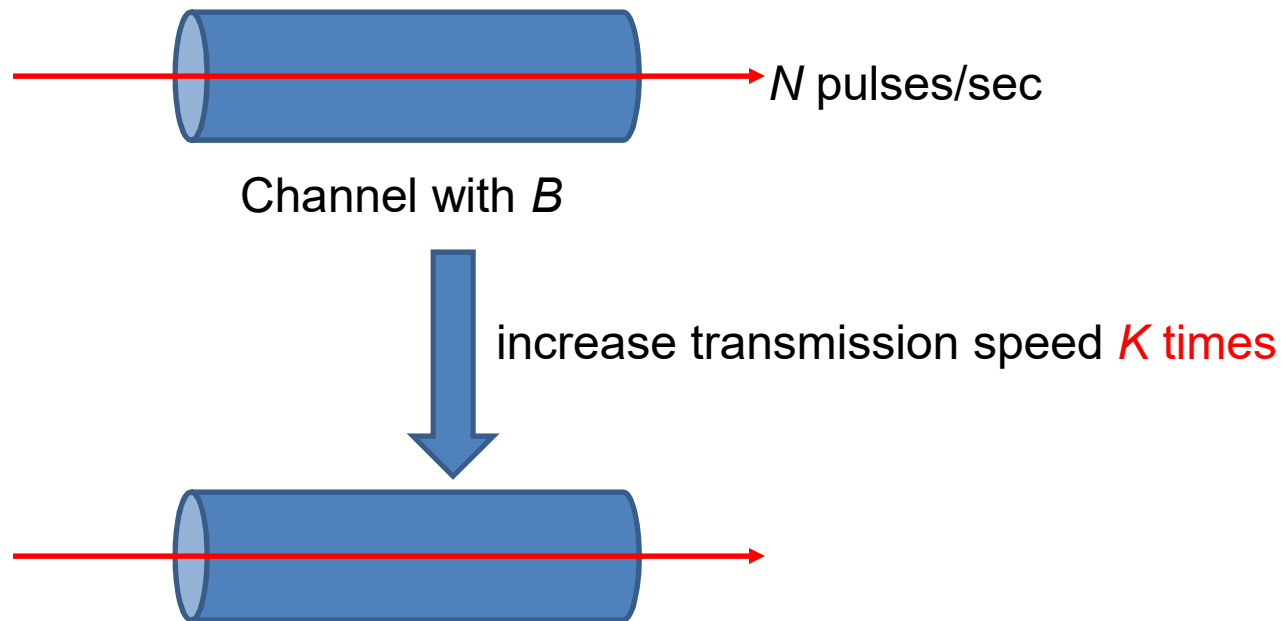
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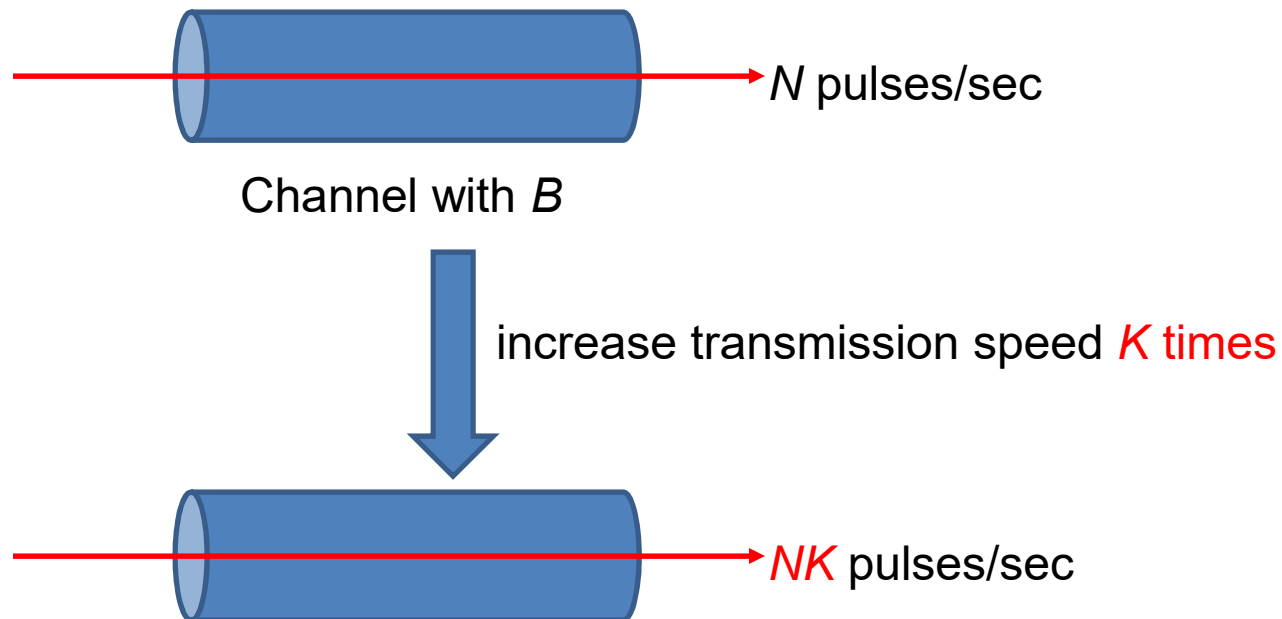
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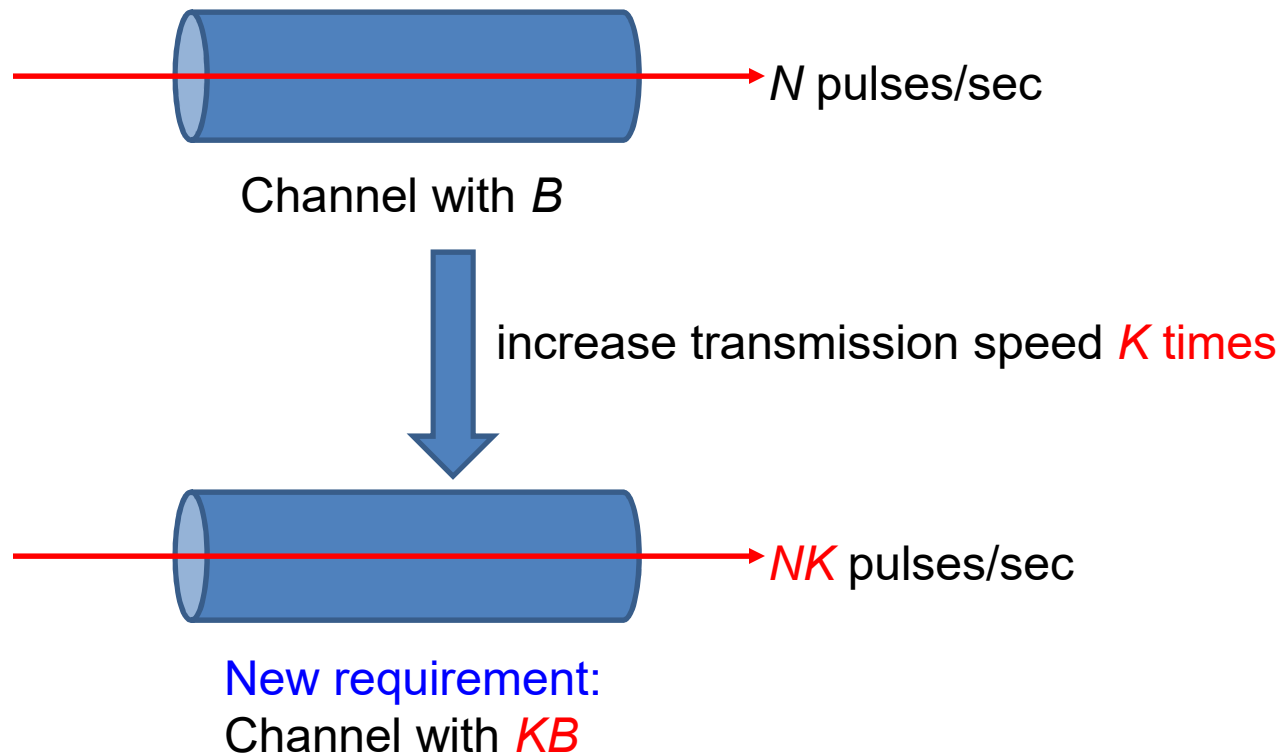
Channel & signal Characteristics

- More transmission speed requires channel with higher bandwidth



Channel & signal Characteristics

- More transmission speed requires channel with higher bandwidth



Channel & signal Characteristics

- Signal Power, P_s
 - Dual role
 - Higher Quality
 - Less channel bandwidth

Channel & signal Characteristics

- Signal Power, P_s for higher quality
 - $P_s++ \rightarrow SNR++$
 - maintains minimum SNR for longer distance
 - Higher SNR means
 - more noise immunity
 - easier detection of pulses

Channel & signal Characteristics

- Signal Power, P_s needs less channel bandwidth
 - Higher signal P_s eases the channel

Channel & signal Characteristics

- Signal Power, P_s needs less channel bandwidth
 - Higher signal P_s eases the channel
 - Shannon's limit on channel capacity

$$C = B \log_2(1 + SNR) \text{ bit/s}$$

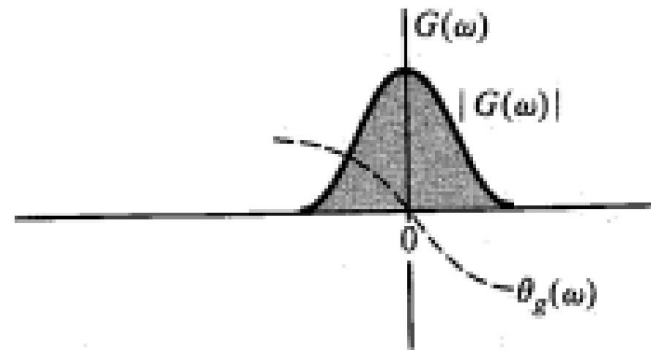
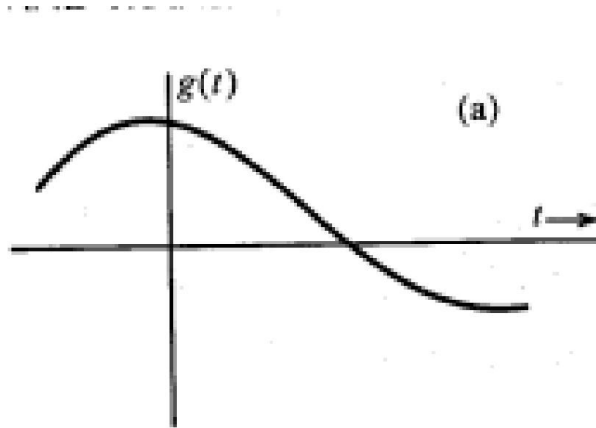
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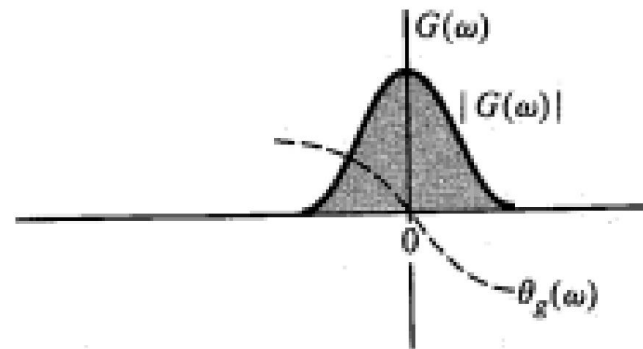
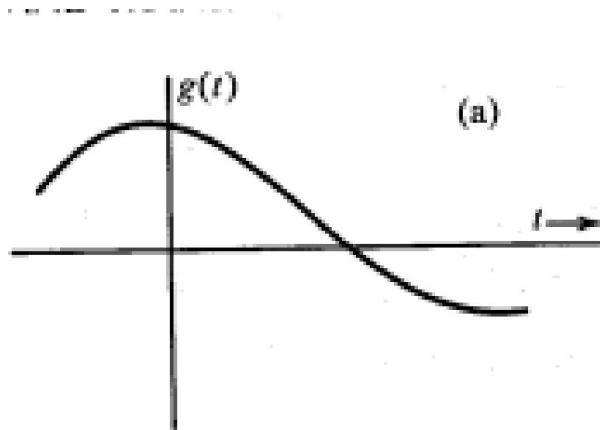
What happens if $SNR = \infty$?

Channel & signal Characteristics

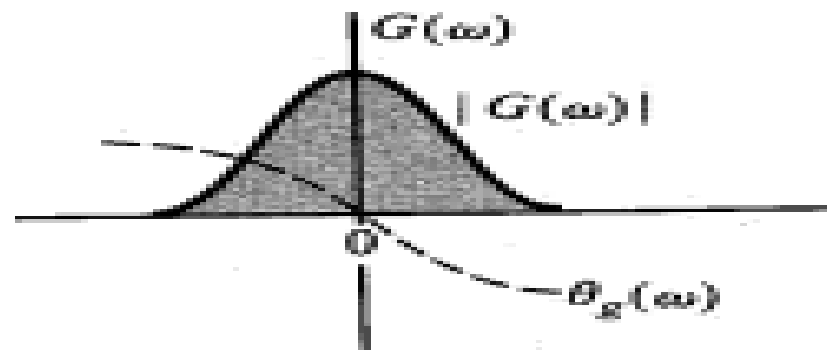
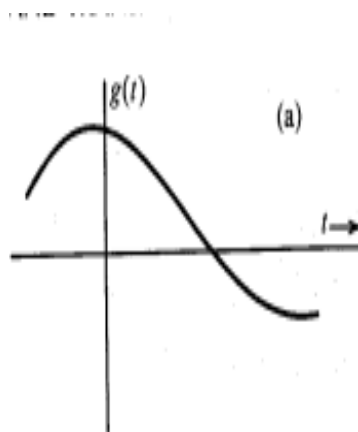


Baseband signal

Channel & signal Characteristics



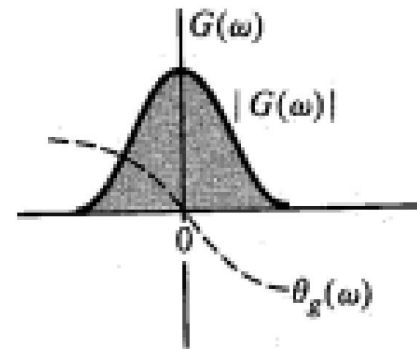
Baseband signal 1



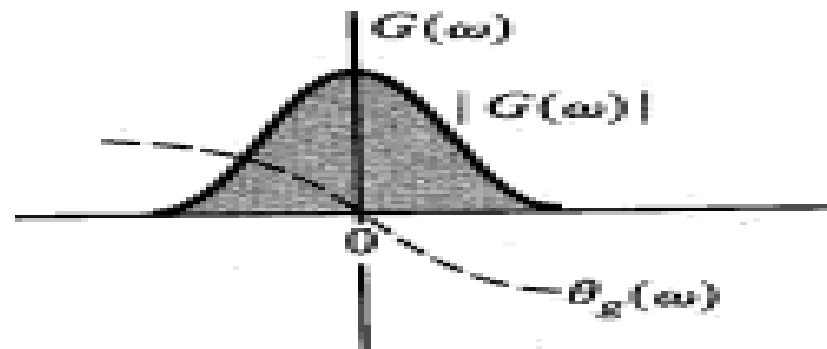
Baseband signal 2

Channel & signal Characteristics

- Assume, these 2 signals need to be transmitted



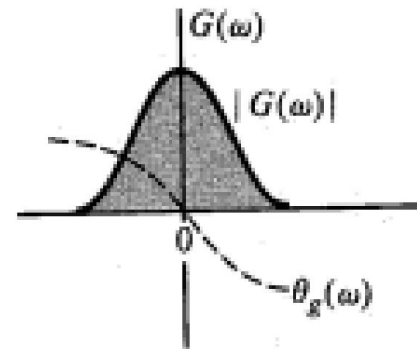
Baseband signal 1



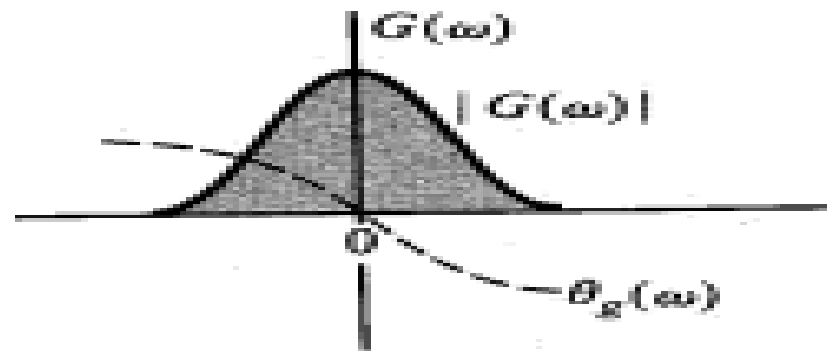
Baseband signal 2

Channel & signal Characteristics

- Assume, these 2 signals need to be transmitted
 - 2 different channels?



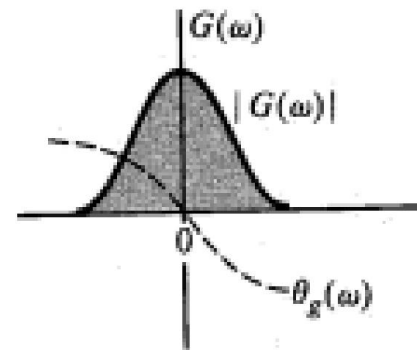
Baseband signal 1



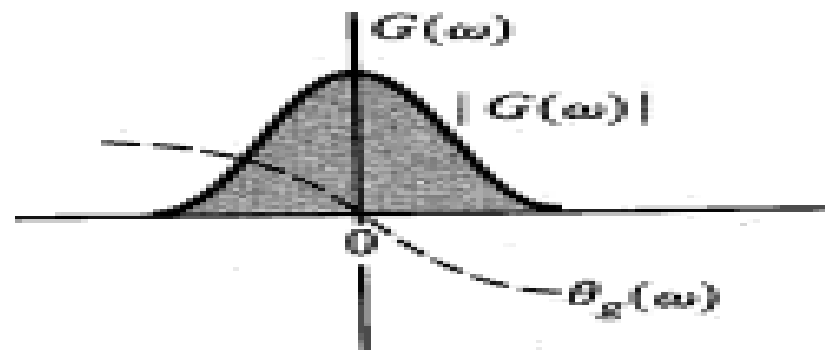
Baseband signal 2

Channel & signal Characteristics

- Assume, these 2 signals need to be transmitted
 - Even, channel bandwidth and signal bandwidth may NOT match!!



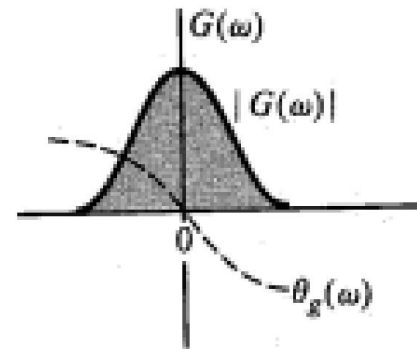
Baseband signal 1



Baseband signal 2

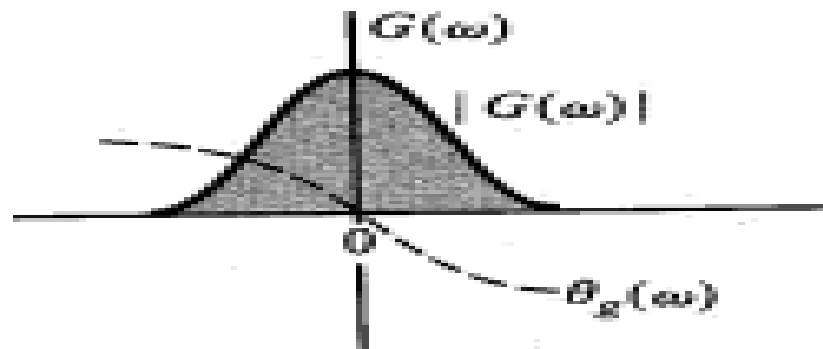
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Baseband signal 1

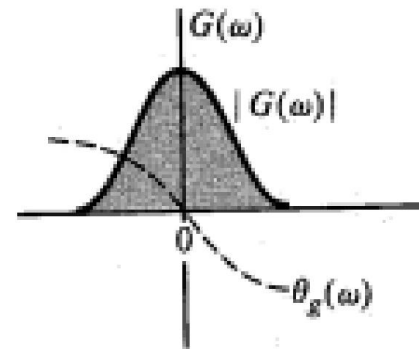
Solution is
modulation



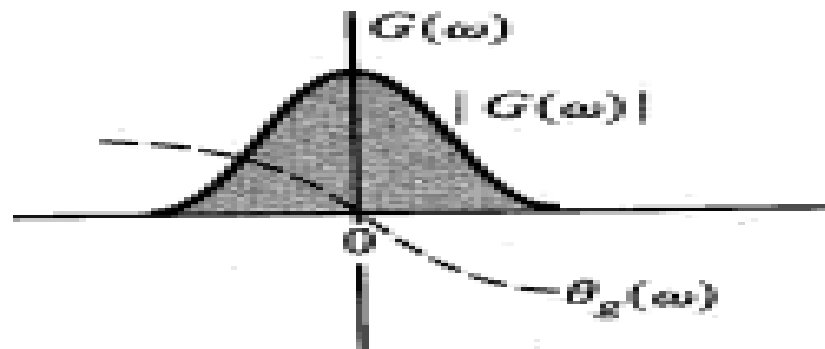
Baseband signal 2

Channel & signal Characteristics: *modulation*

- Use carrier signal to shift these 2 signals to different frequency positions



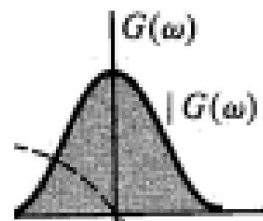
Baseband signal 1



Baseband signal 2

Channel & signal Characteristics: *modulation*

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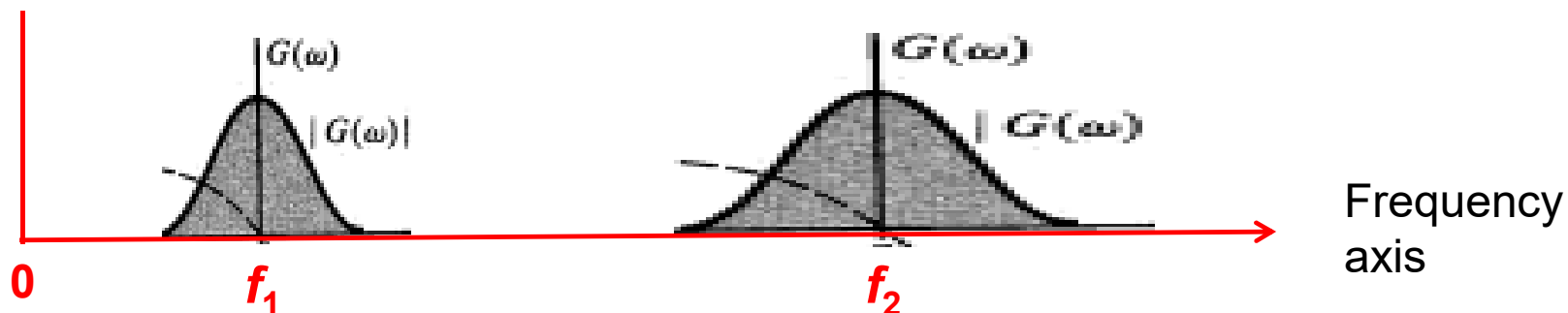


Baseband
signal 1



Baseband
signal 2

After
modulation:

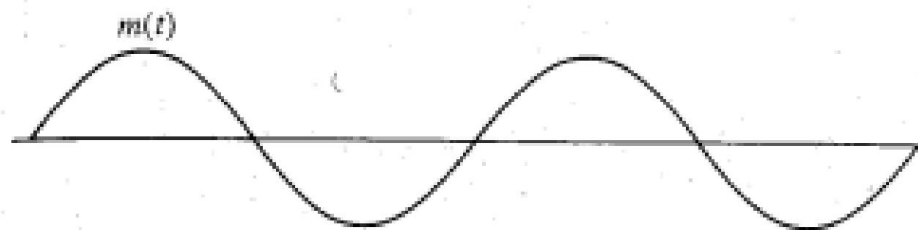


Channel & signal

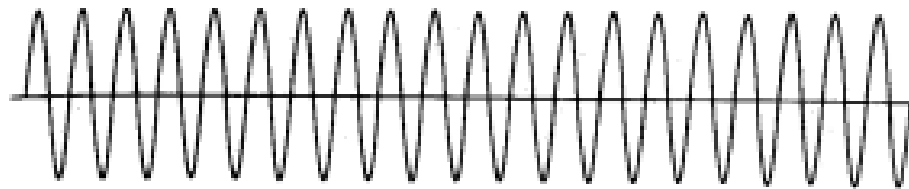
Characteristics: *modulation*

- other examples of modulation
(shown in time domain)

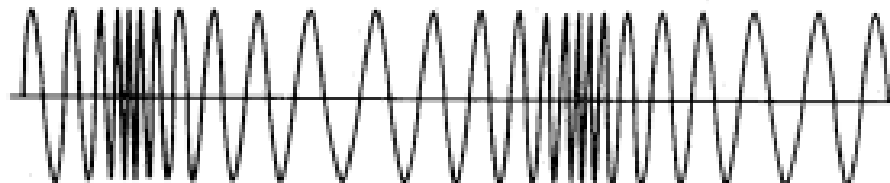
Baseband
signal



Carrier signal



Modulated
signal

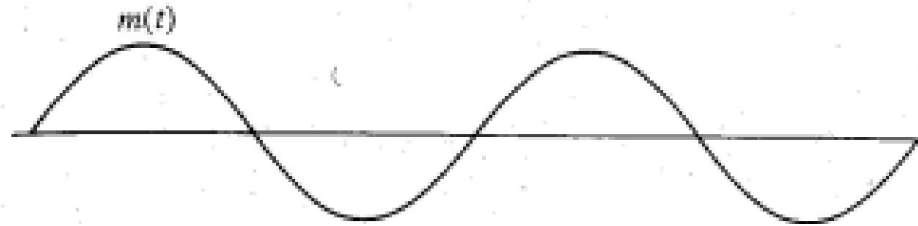


Channel & signal

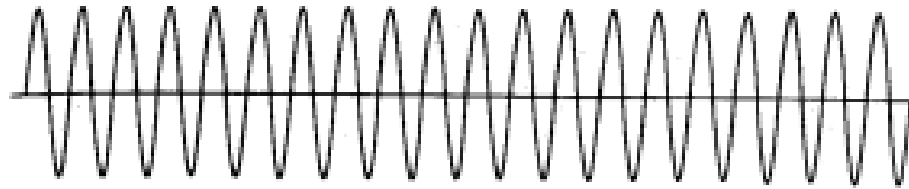
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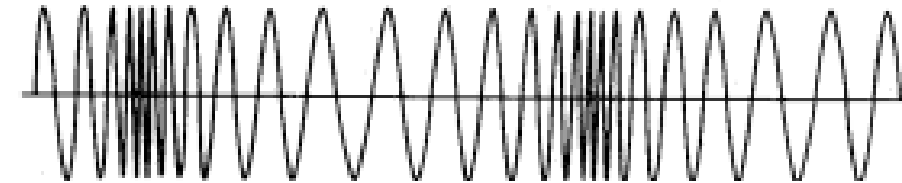
Baseband
signal



Carrier signal



Modulated
signal



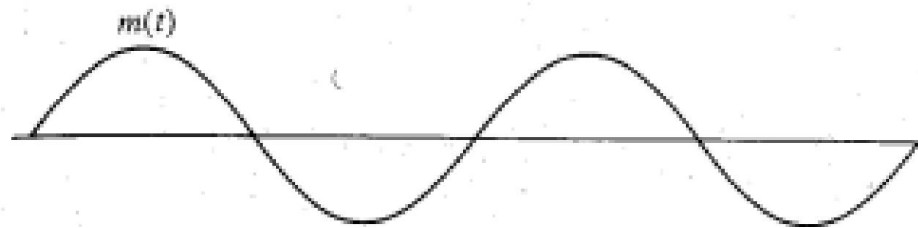
**Frequency
modulation
(FM)**

Channel & signal

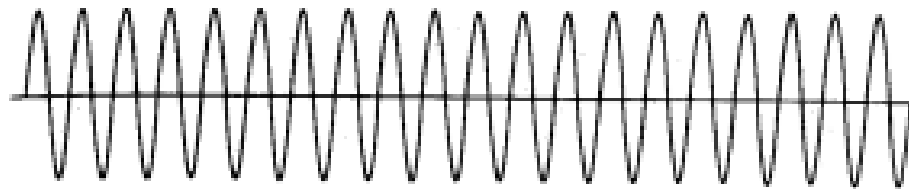
Characteristics: *modulation*

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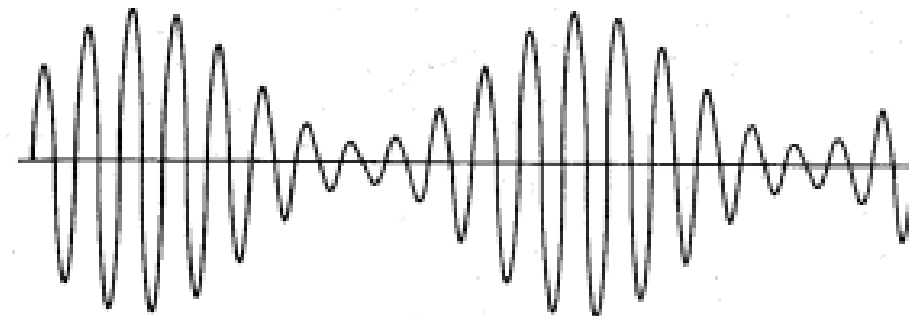
Baseband
signal



Carrier signal



Modulated
signal



**Amplitude
modulation
(AM)**

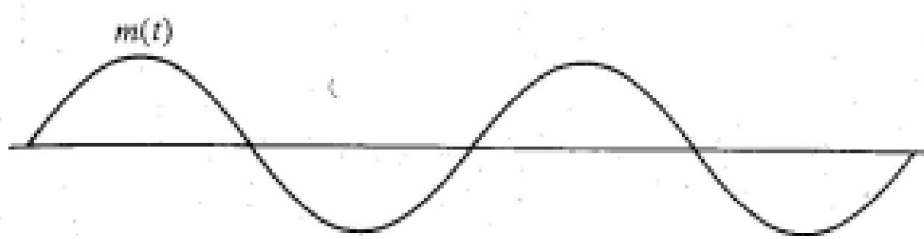
Channel & signal

Characteristics: *modulation*

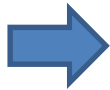
- Phase modulation (PM)
- Changes phase angle of the signal

Channel & signal Characteristics: *modulation*

Other reasons for modulation



Low frequency



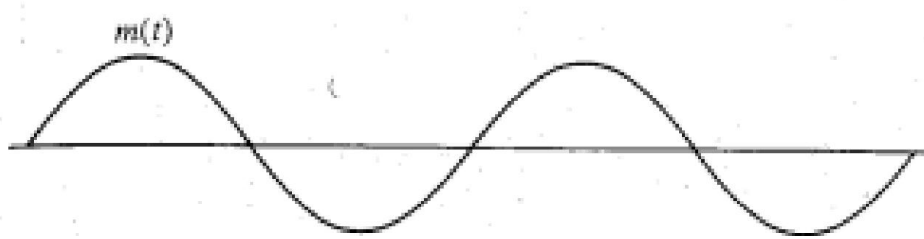
Higher
wavelength



Generating
antenna requires
higher dimension

Channel & signal Characteristics: *modulation*

Other reasons for modulation

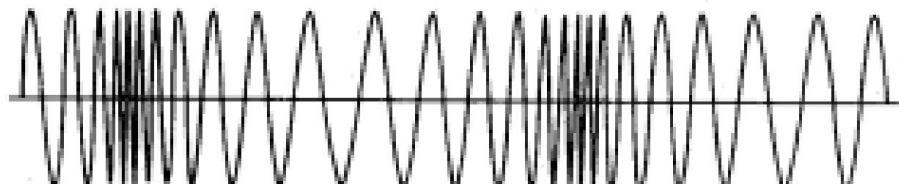


Low frequency → Higher wavelength → Generating antenna requires higher dimension

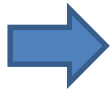
- Speech signal characteristics
 - Frequency range: 100 – 3000 Hz
 - Wavelength 100 to 3000 km
 - requires *impractically* large antenna

Channel & signal Characteristics: *modulation*

Other reasons for modulation



Modulating a High
frequency carrier



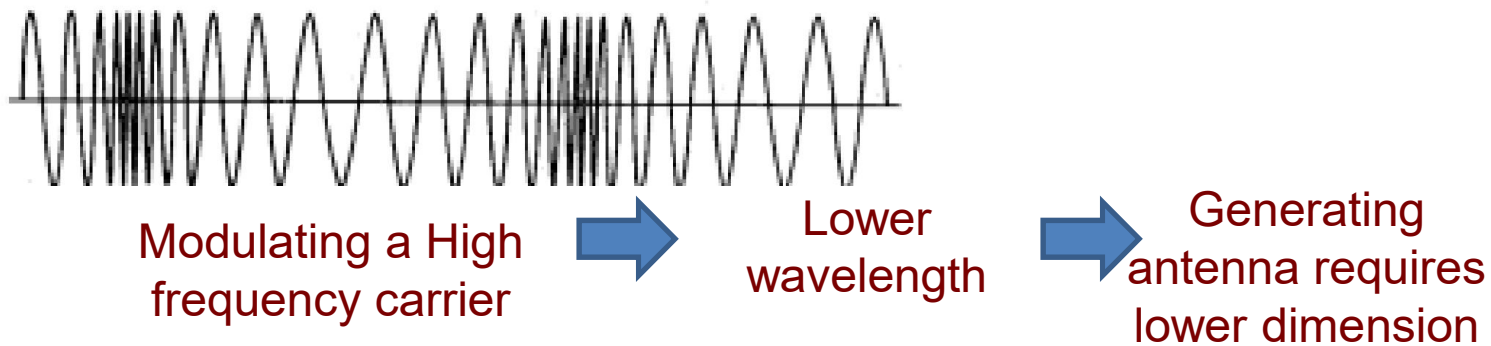
Lower
wavelength



Generating
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lower dimension

Channel & signal Characteristics: *modulation*

Other reasons for modulation



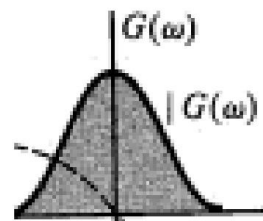
- 10 MHz carrier signal characteristics
 - Wavelength 30 m
 - requires antenna of size 3m

Channel & signal

Characteristics: *modulation*

- Use carrier signal to shift these 2 signals in different frequency positions

recall this example!

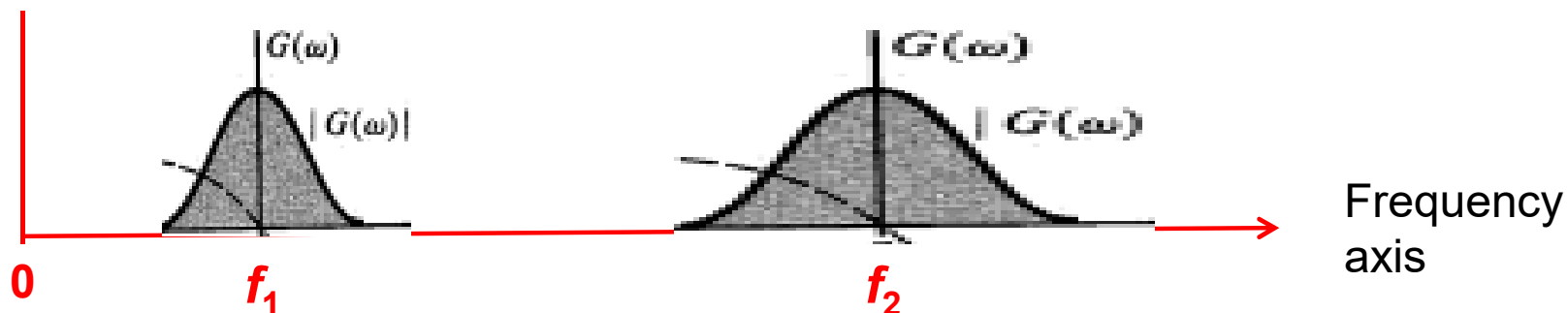


Baseband
signal 1



Baseband
signal 2

After
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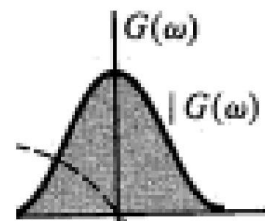


Channel & signal Characteristics: *modulation*

- Use carrier signal to shift these
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positions

also called
*Frequency division
multiplexing (FDM)*

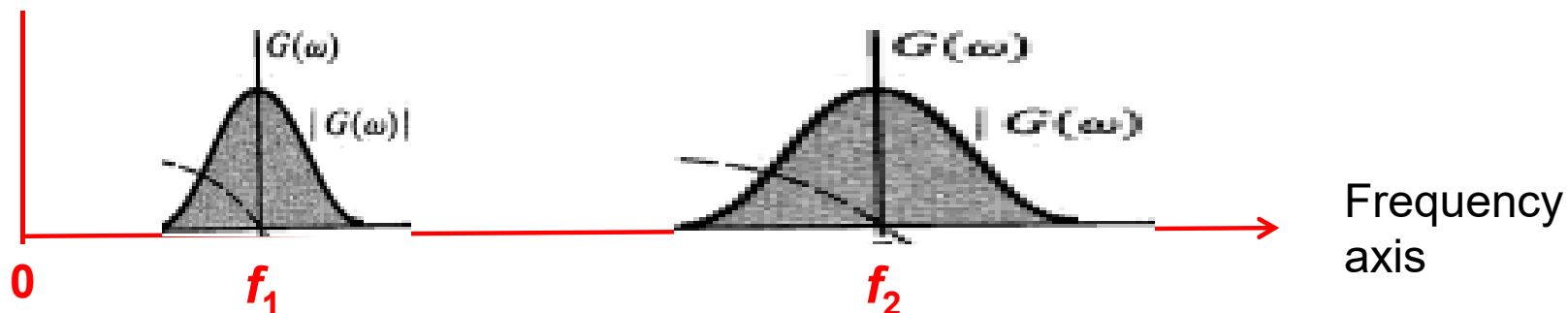
After
modulation:



Baseband
signal 1



Baseband
signal 2

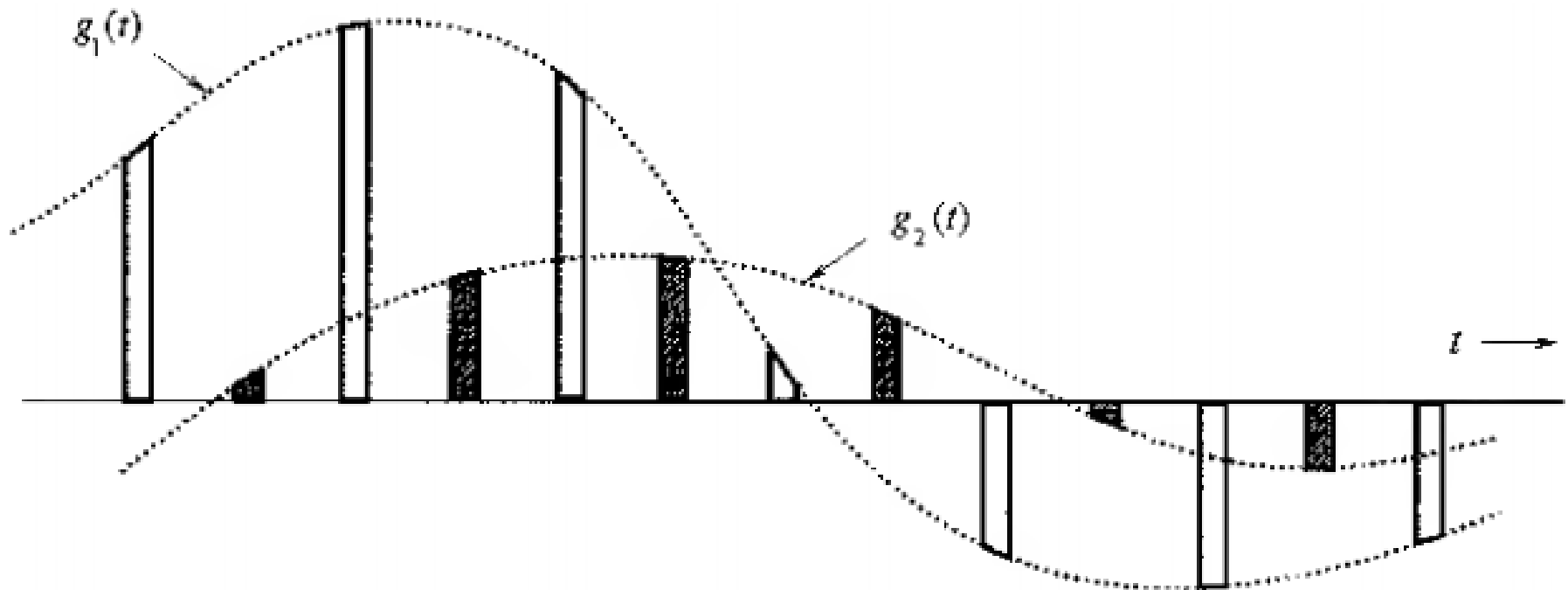


Channel & signal Characteristics: *modulation*

- Time division multiplexing (TDM)
 - Interleave pulses from different signals in time domain signal

Channel & signal Characteristics: *modulation*

- Time division multiplexing (TDM)
 - Interleave pulses from different signals in time domain signal



Channel & signal

Characteristics: *DeModulation*

- Done at the receiving end
 - Bandpass filter separates appropriate signal
 - Makes necessary corrections for amplitude, frequency and phase changes